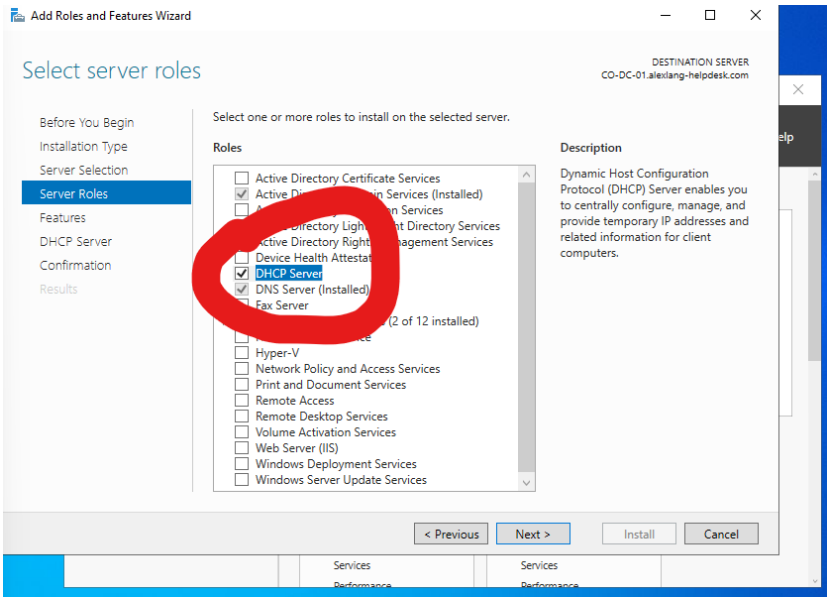
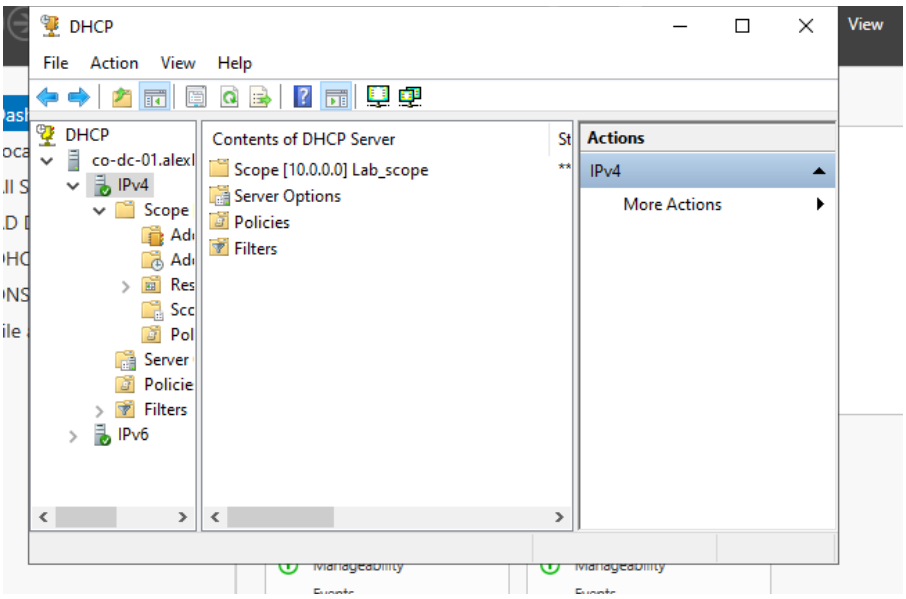
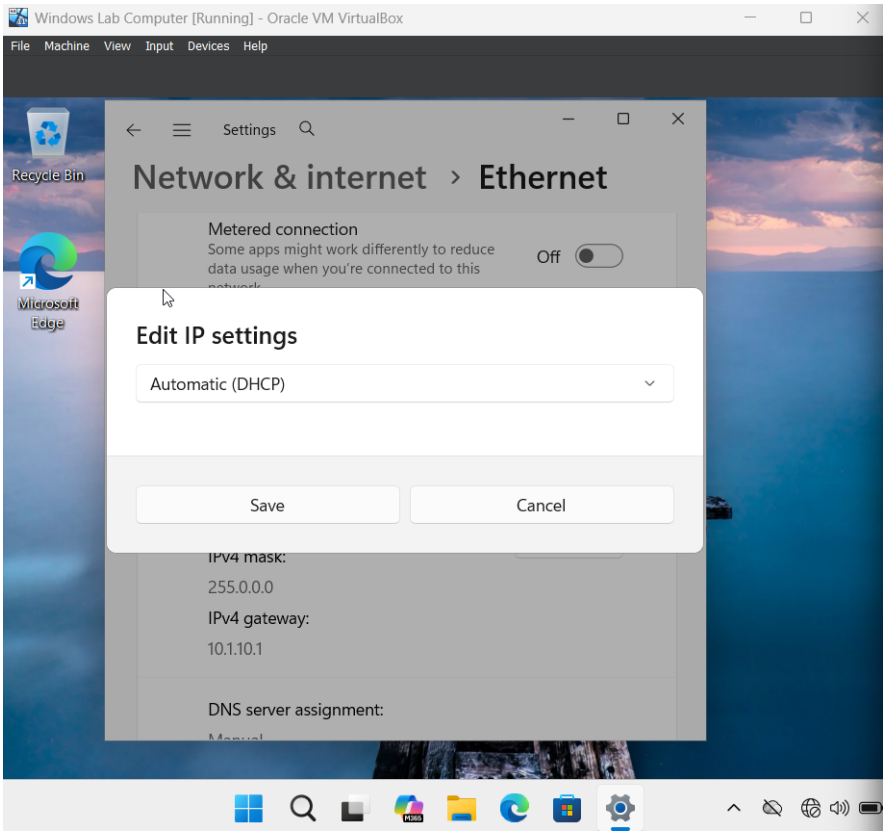
Day 11 Setting up DHCP Server



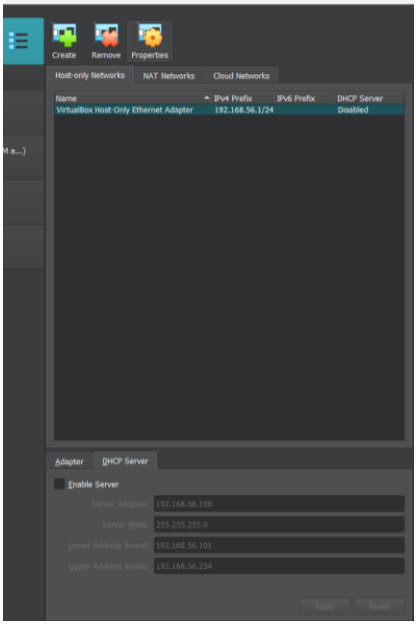
Installing DHCP Server incside of the add roles and features wizard



DHCP custom scope setup, using a custom IP address range, excludes 10.1.10.2 which is my servers static IP, added a lease of 10 days, and used 10.1.10.1 as my default gateway (which is my servers static IP).

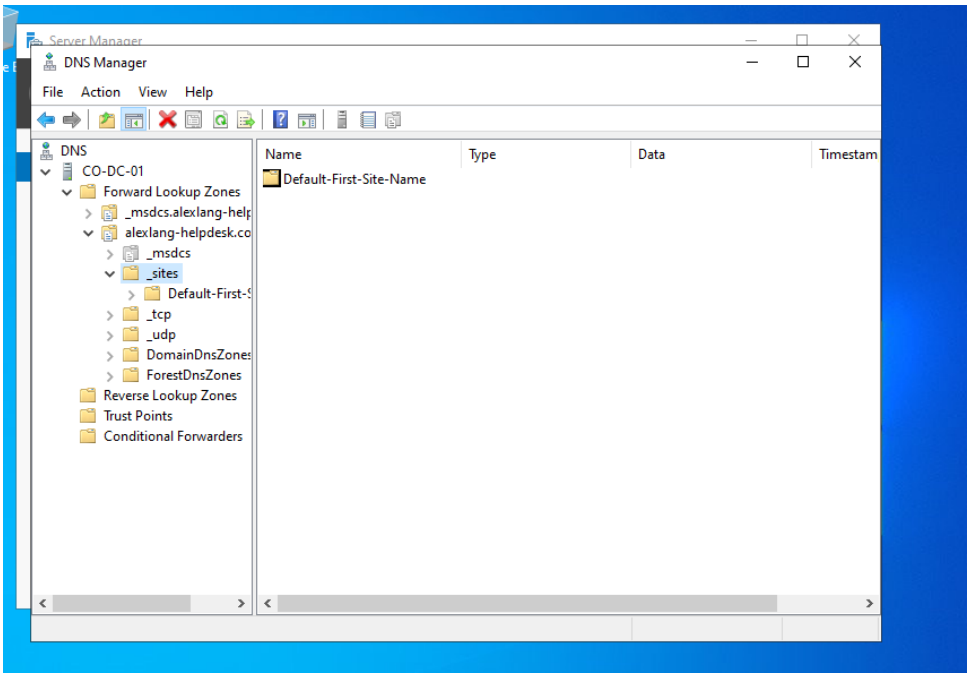


From here, I switched over to my lab computer, went into Network/Internet settings, and changed IP settings to use DHCP instead of static IP assignment.

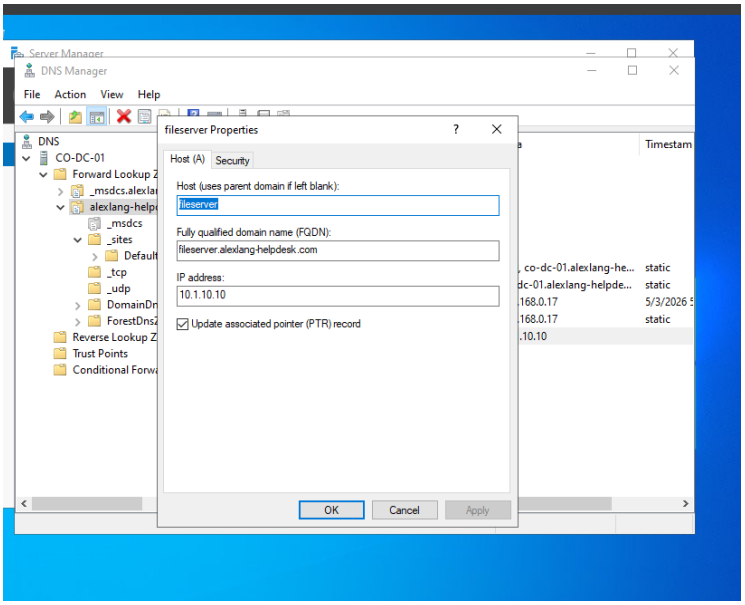


Went into the hypervisor to disable the DHCP Server since we just created one

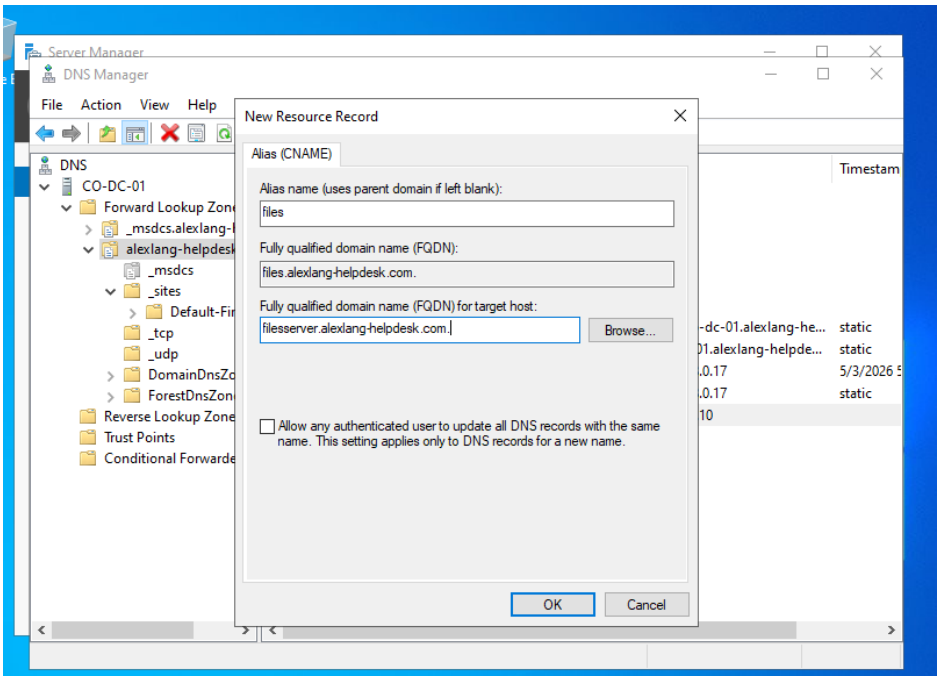
Day 12 DNS Troubleshooting



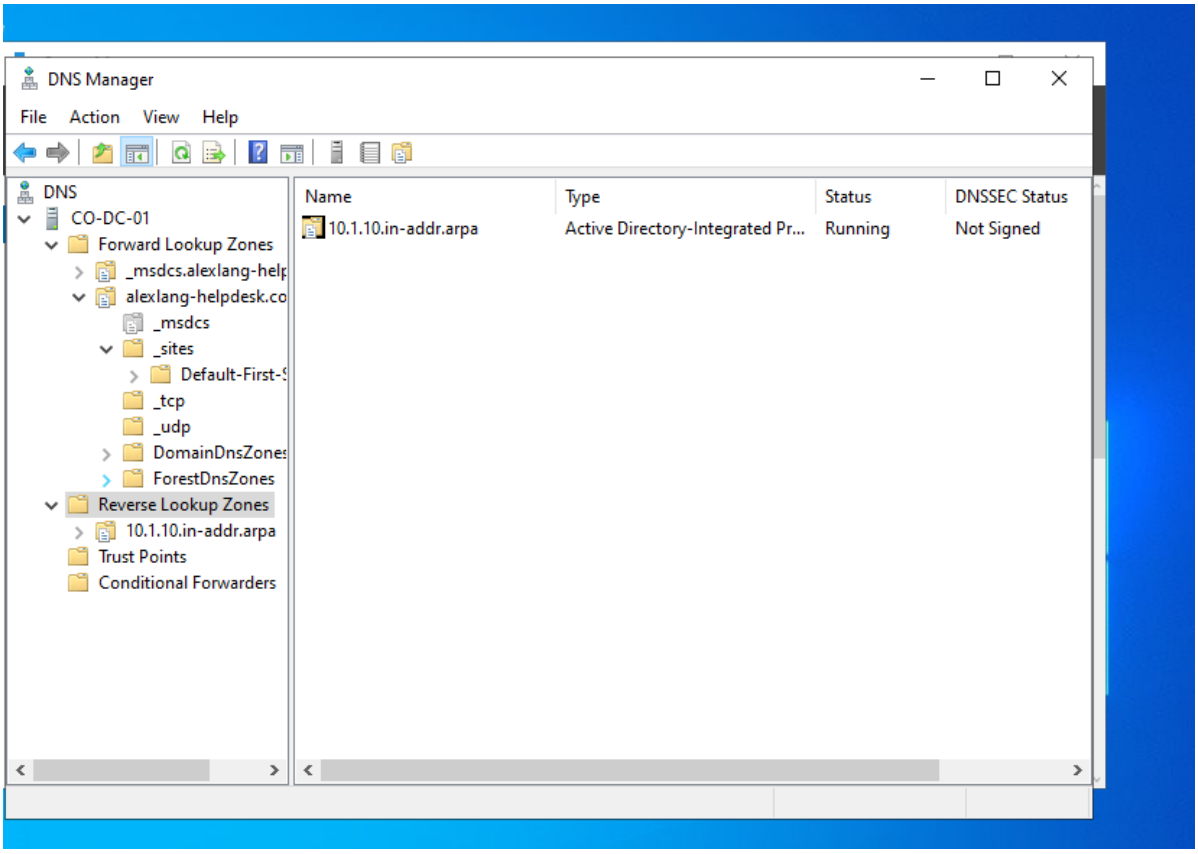
Took some time to examine all the DNS records that were created automatically by AD. Seeing these SOA, NS, A, and SRV records was really refreshing becuase I learned what each one does while studying for CompTIA's A+.



Created an A record

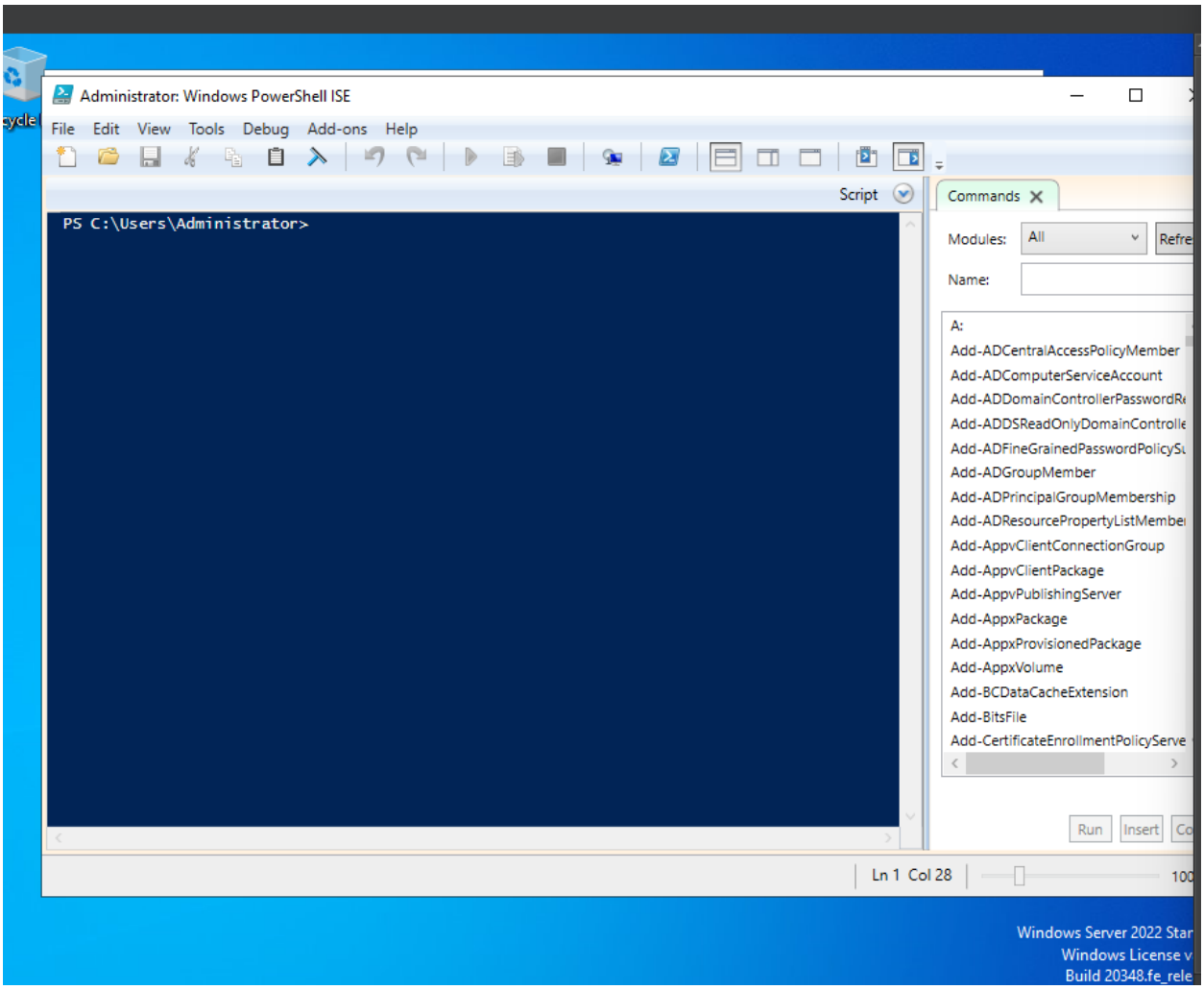


Created a CNAME record also known as an alias for the server just created

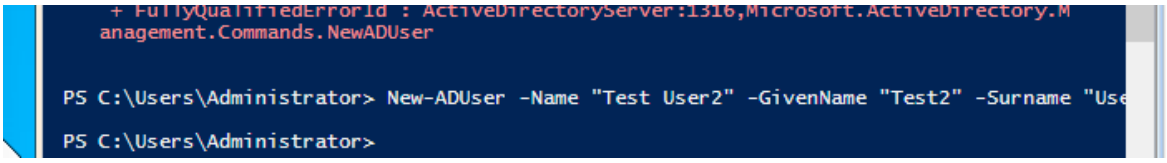


Created a reverse lookup zone

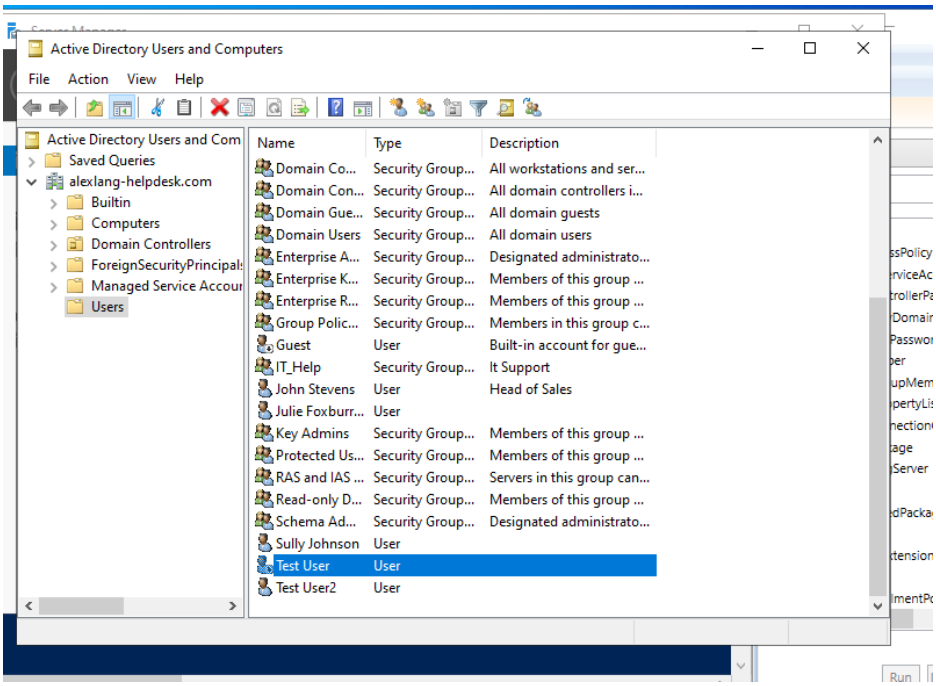
Day 13 PowerShell ISE



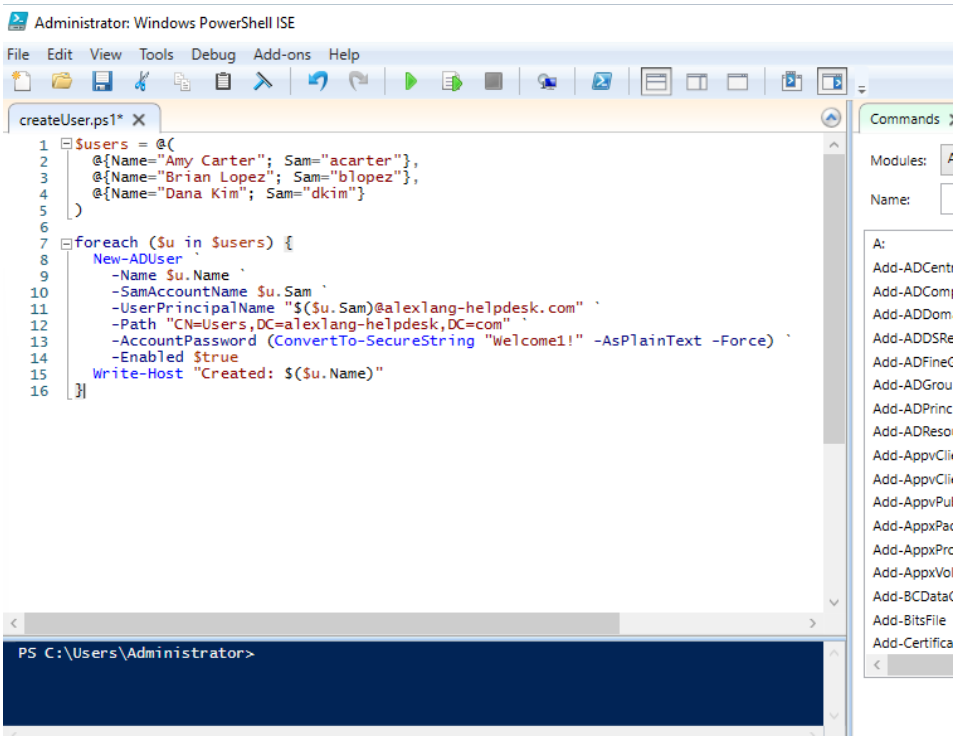
Going to write various shell scripts to automate tasks inside of AD



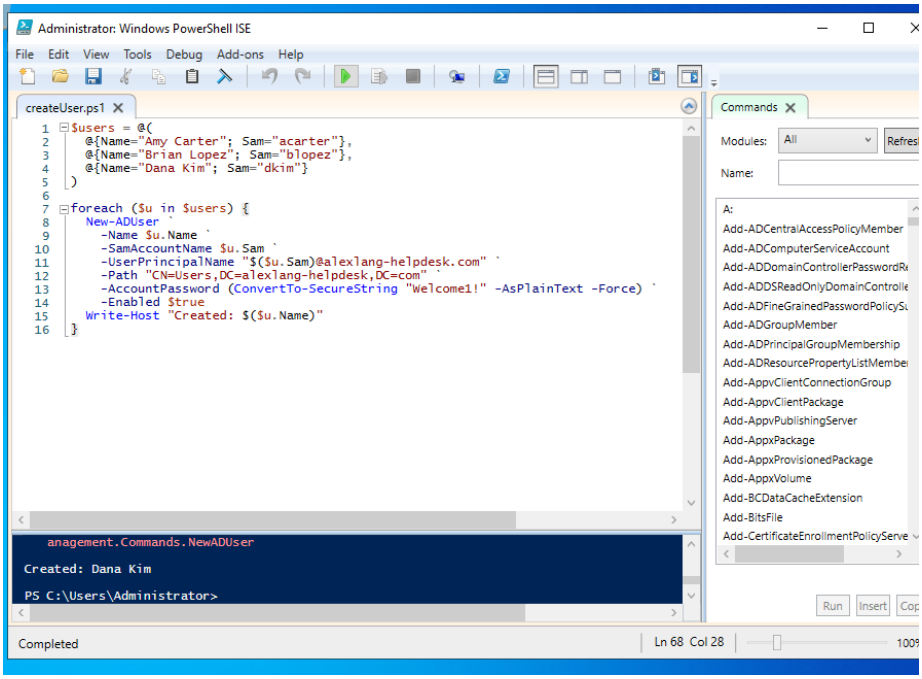
Automated the process of creating user accounts using AD specifications. I did use AI to help me write these scripts for time purposes, and to help me get a better understanding of PowerShell ISE documentation.



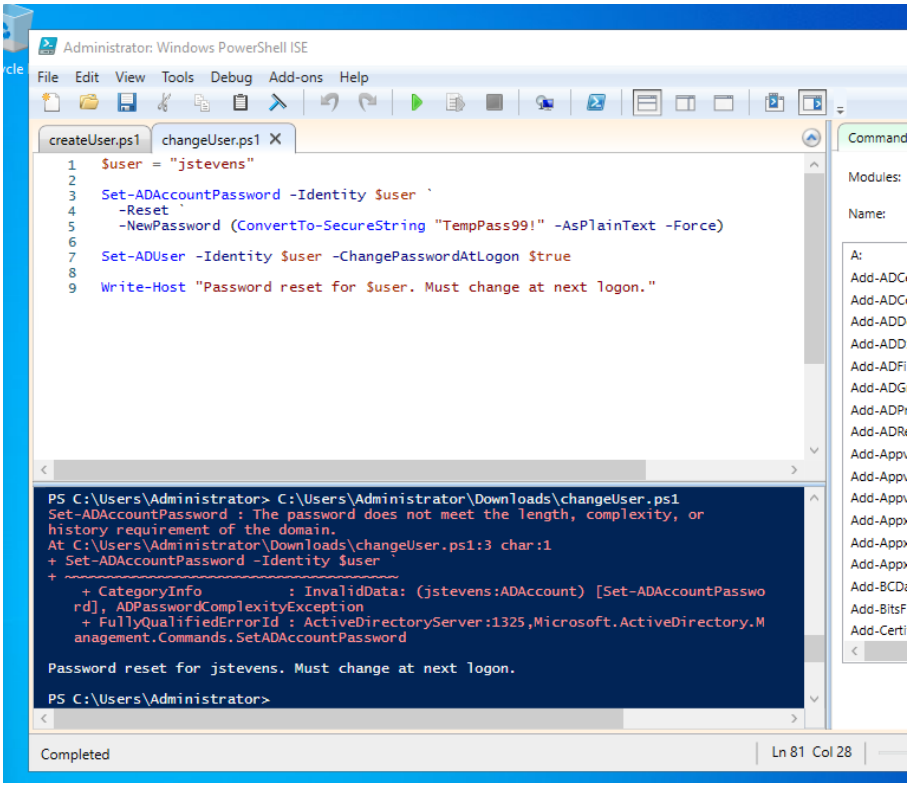
Test users 1 and 2 were successfully created



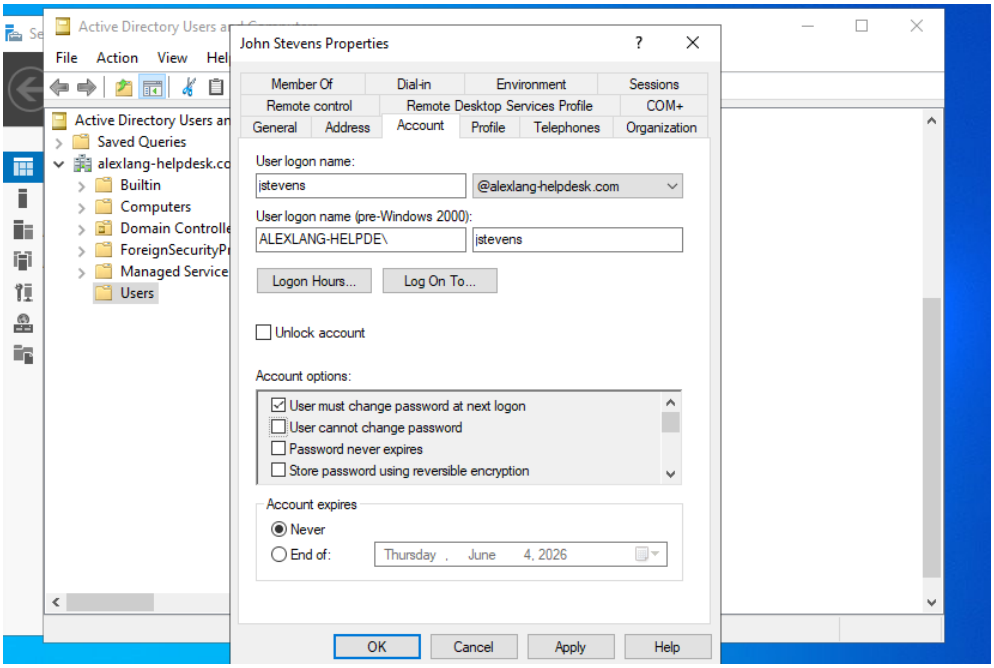
Created a .ps1 file so that the creation of user accounts can be automated a lot easier



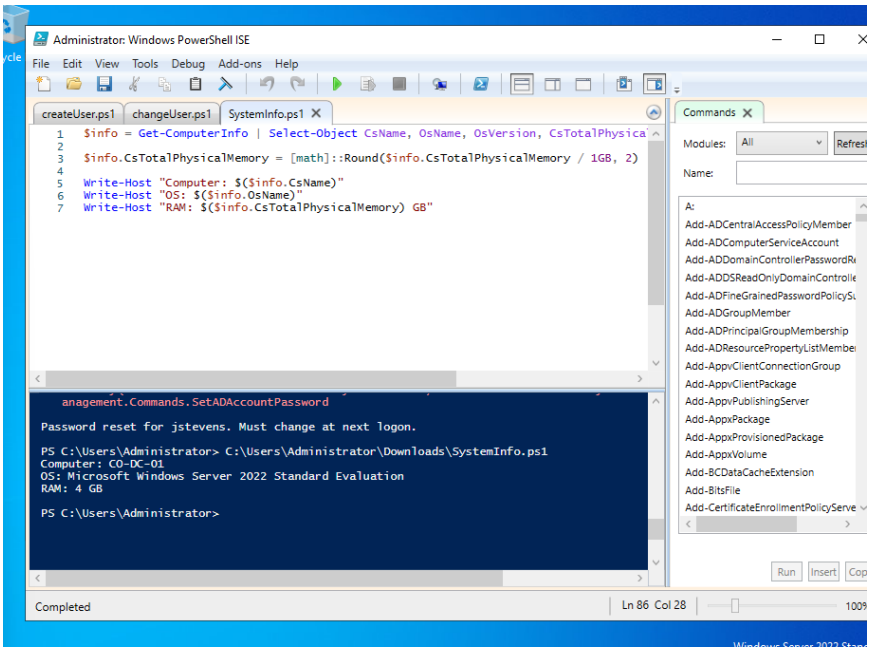
Showing the creation of given user accounts



Wrote a script for password reset as well, now user jstevens must reset there password next time they try to login to thier workstation

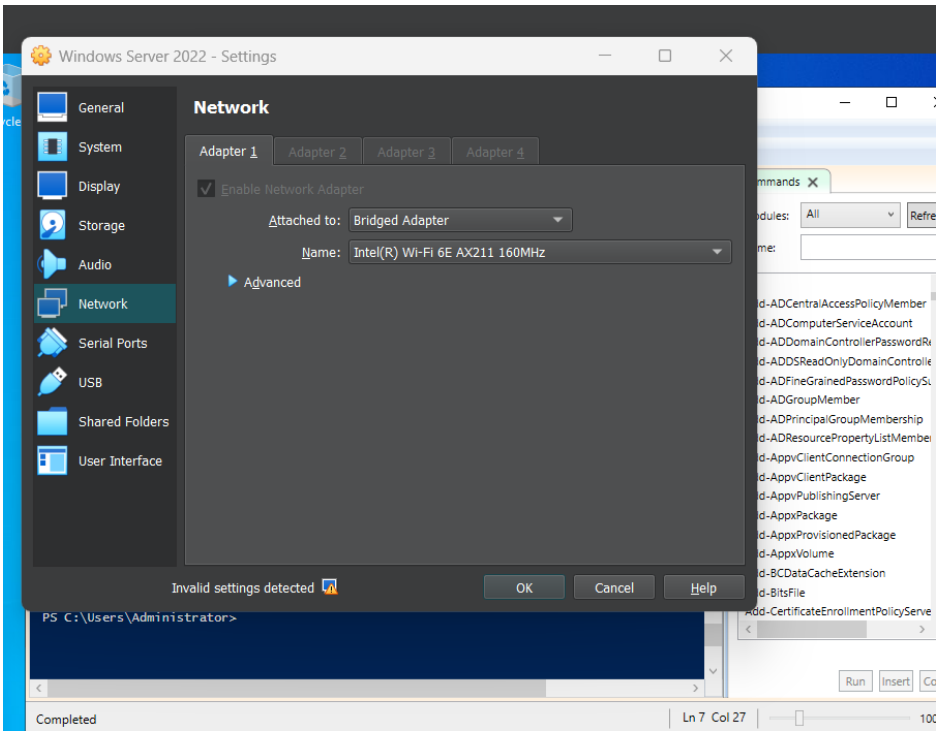


Navigating back to AD Users and Groups, it clearly said that the user must change there login meaning the script ran successfully

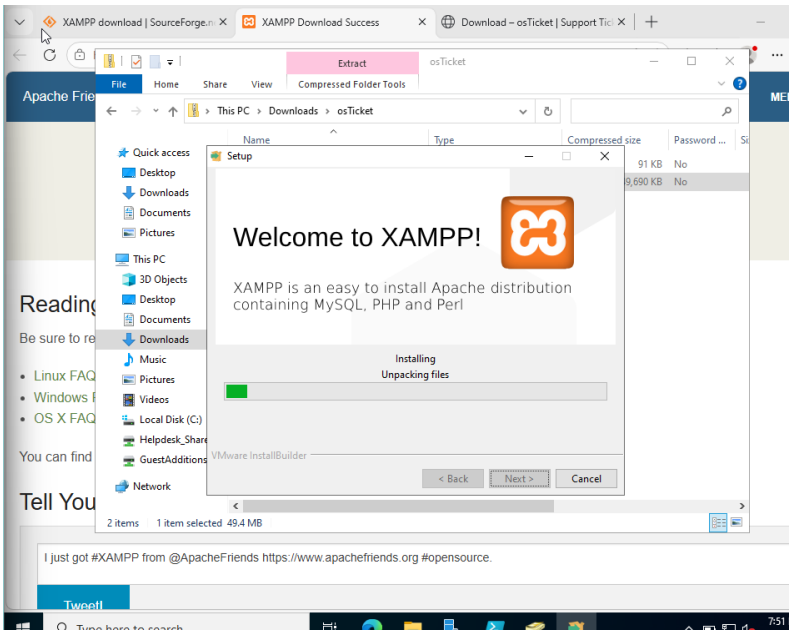


Wrote one more to get some basic info about the system

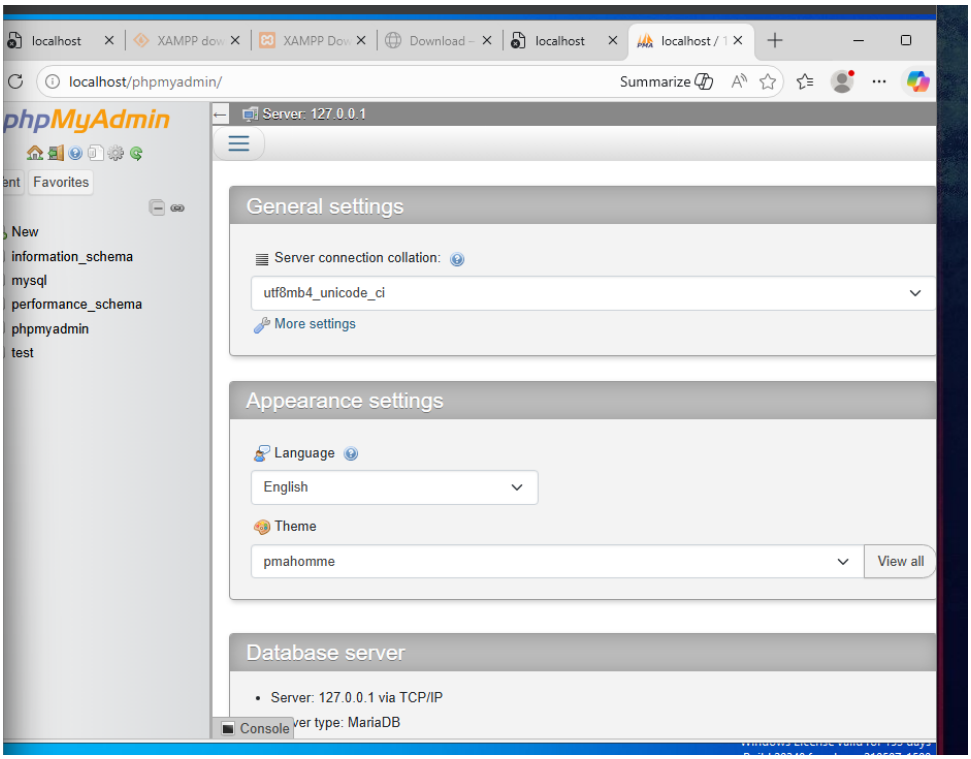
Day 14 Ticketing System with OS Ticket



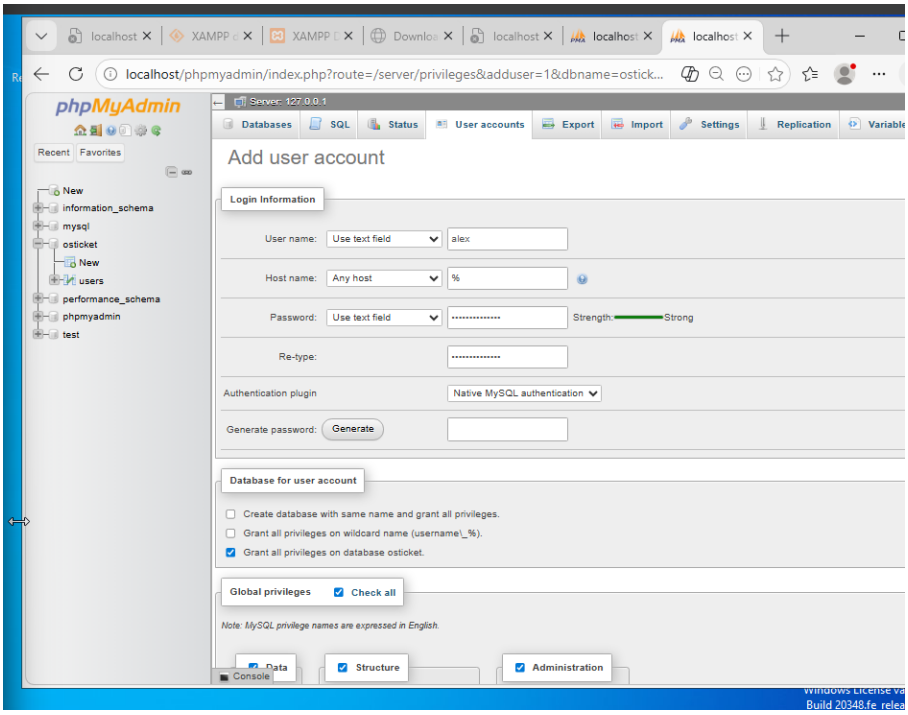
During most of this experiment so far my VM has been on a host-only network connection but I need a network connection to install os ticket so temporally making it a bridge adapter



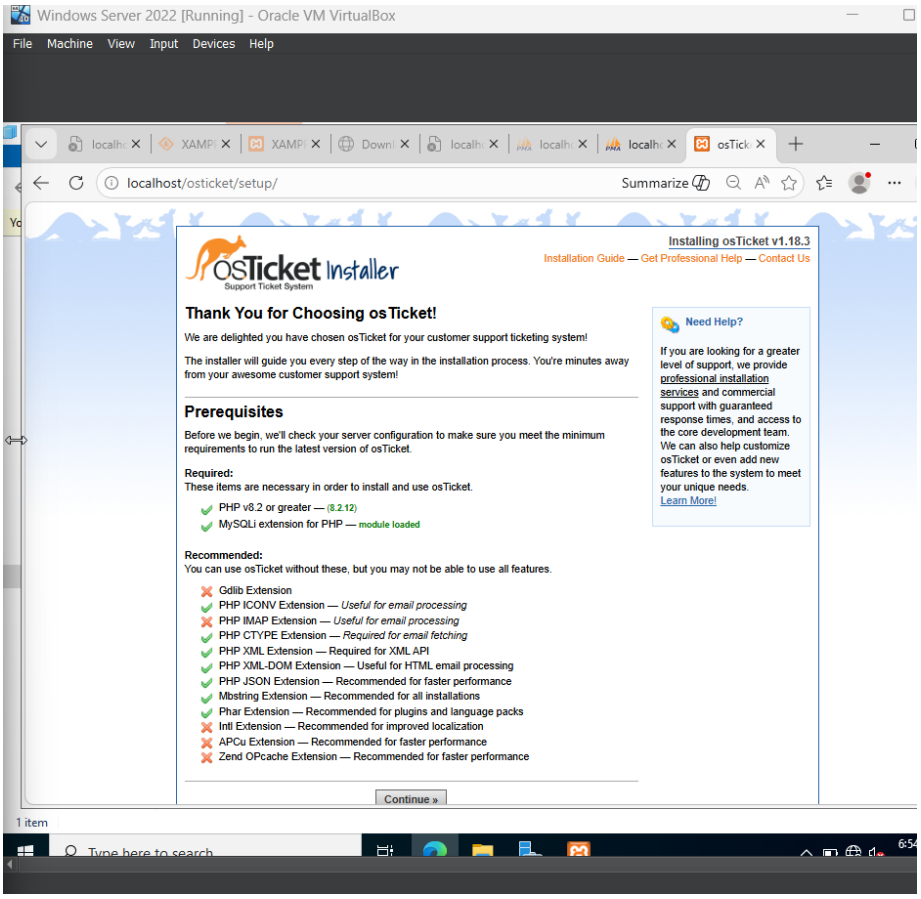
Installed XAMPP(Apache) so I have a local webserver that os ticket can run on and then installed osTicket on the XAMPP server.



XAMPP Control pannel successfully configured, and phpMyAdmin successfully loaded

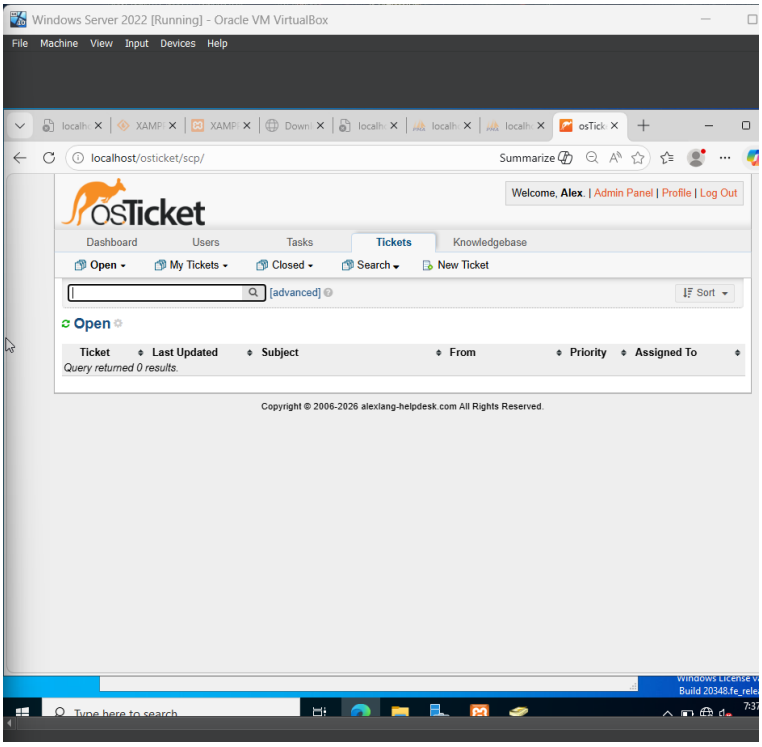


Created a database, added a user account, and gave that account admin permissions

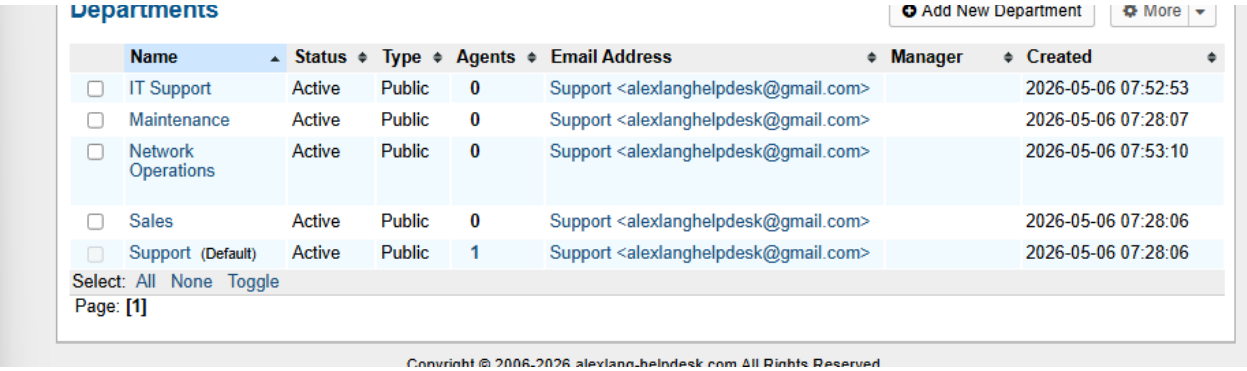


Had to some around some files around and rename some folders, but finally got setup configured in the right location

Updated the config file so that the server could find it



OSTicket dashboard



Departments

Add New Department | More

	Name	Status	Type	Agents	Email Address	Manager	Created
☐	IT Support	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:52:53
☐	Maintenance	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:28:07
☐	Network Operations	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:53:10
☐	Sales	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:28:06
	Support (Default)	Active	Public	1	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:28:06

Select: All None Toggle

Page: [1]

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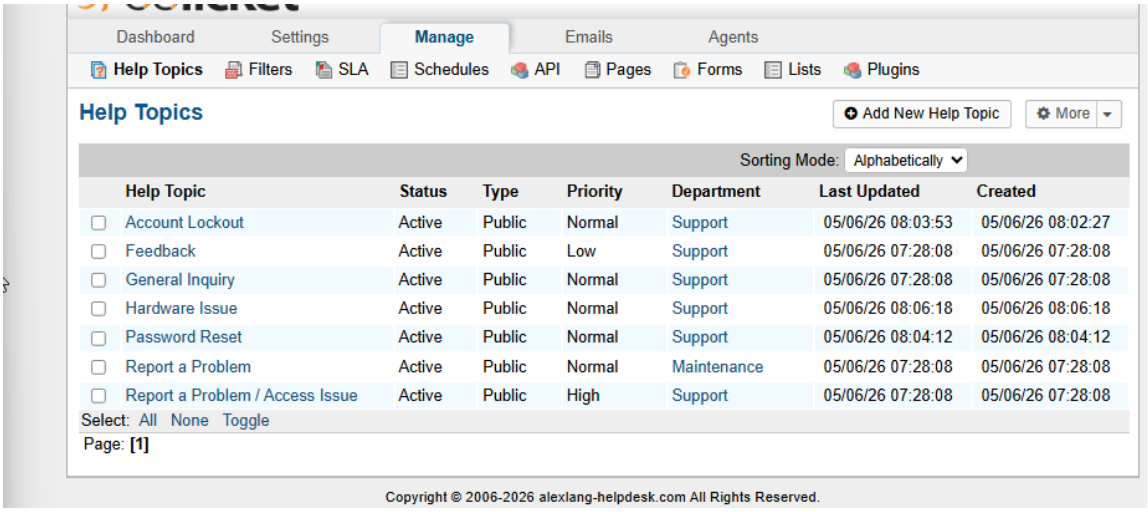
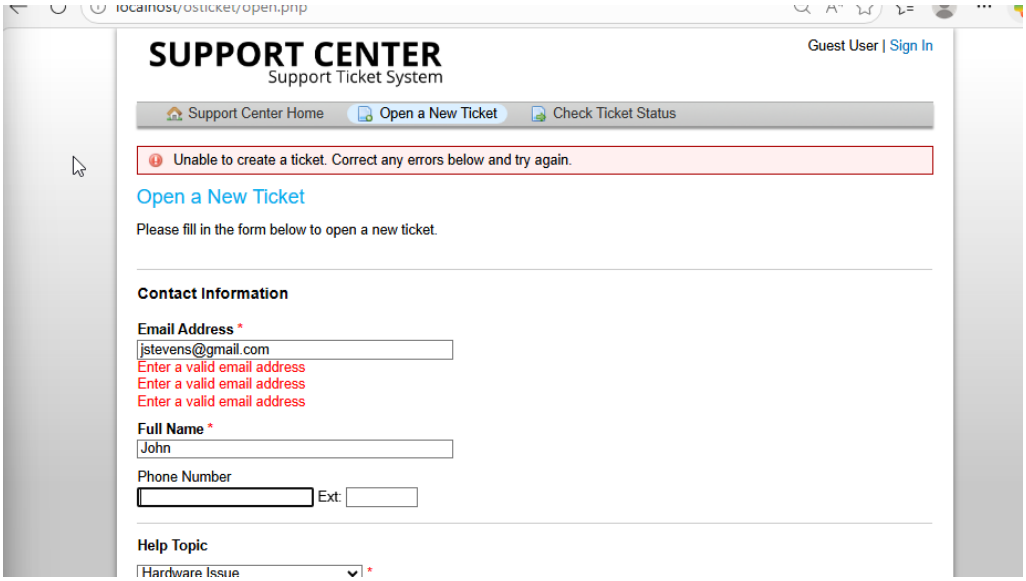
Created 2 new departments, IT Support and Network Operations



Created some general new help topics



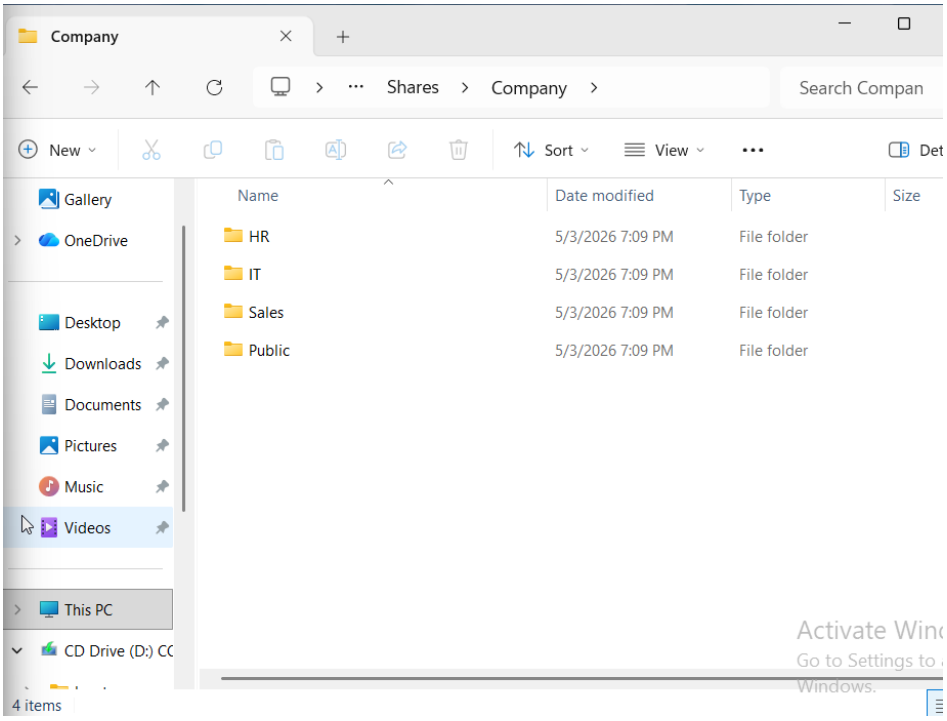
Had a problem with the formal of the emails I was using to create tickets, so I had to navigate back to the admin panel and uncheck the setting “Require email verification on “Require email verification on ticket status”. That also did not work right awat, and it would only accept tickets that.

I got stuck here, and couldnt add any further tickets, will try again later

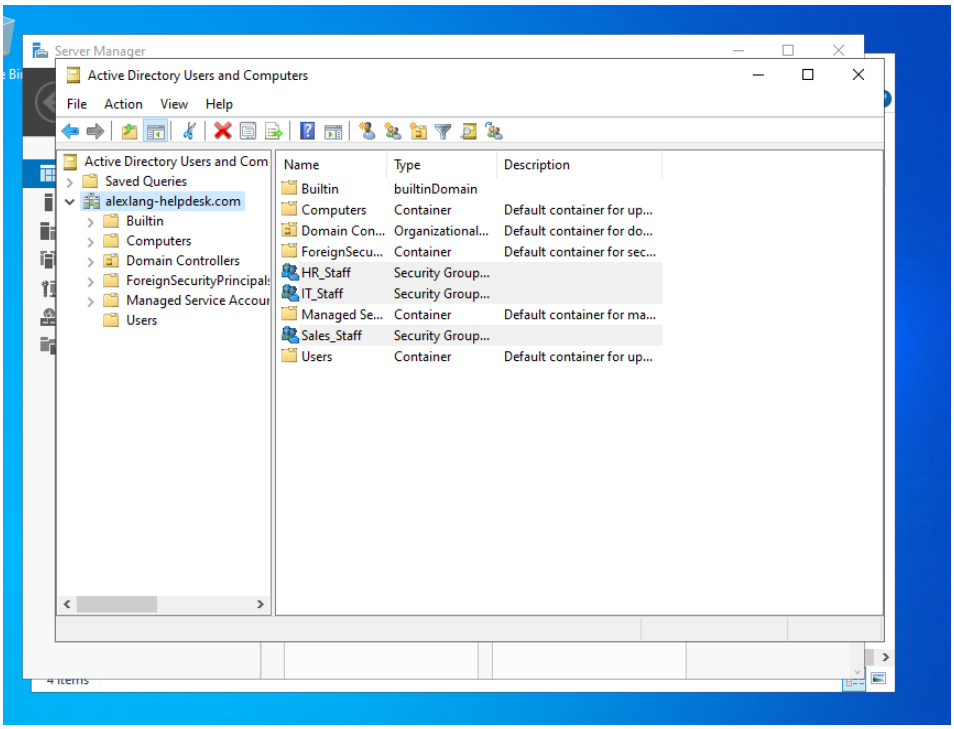
Observations/Results

- I had issues initially setting up the configuration file because the server could not find it (had to do some file configuration management to fix the location issue)
- I also had issues after trying to install osTicket when the server could find the config file and kept getting a permissions error (The configure file admin did not have the proper permissions)

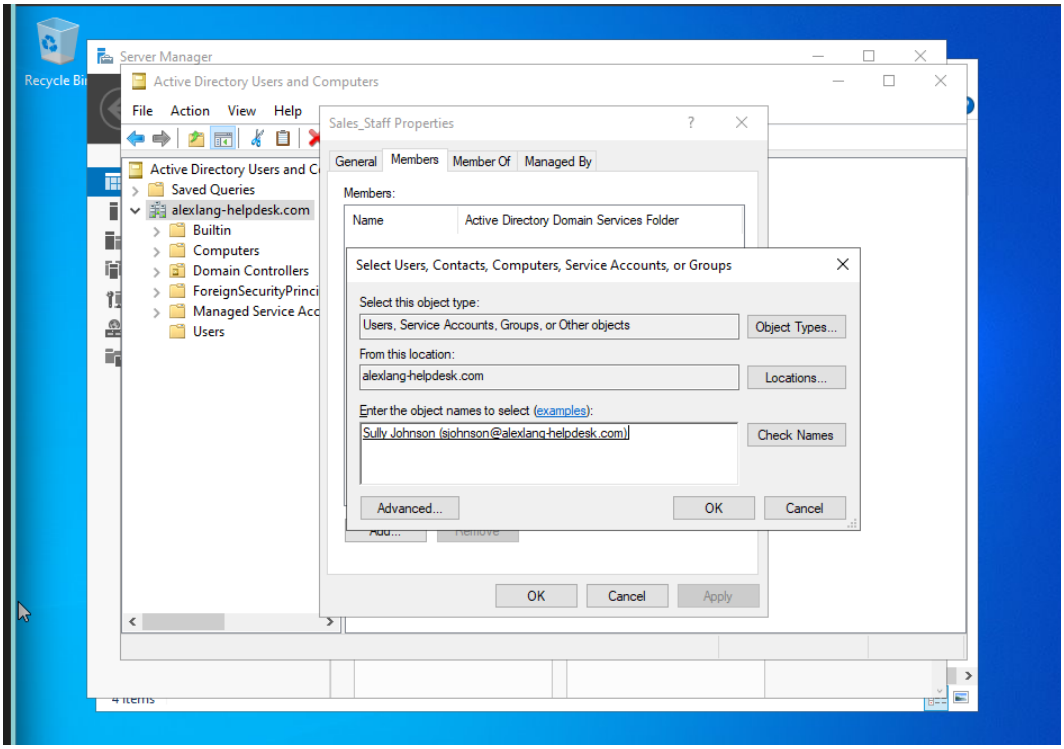
Day 15 NTFS Permissions and Security



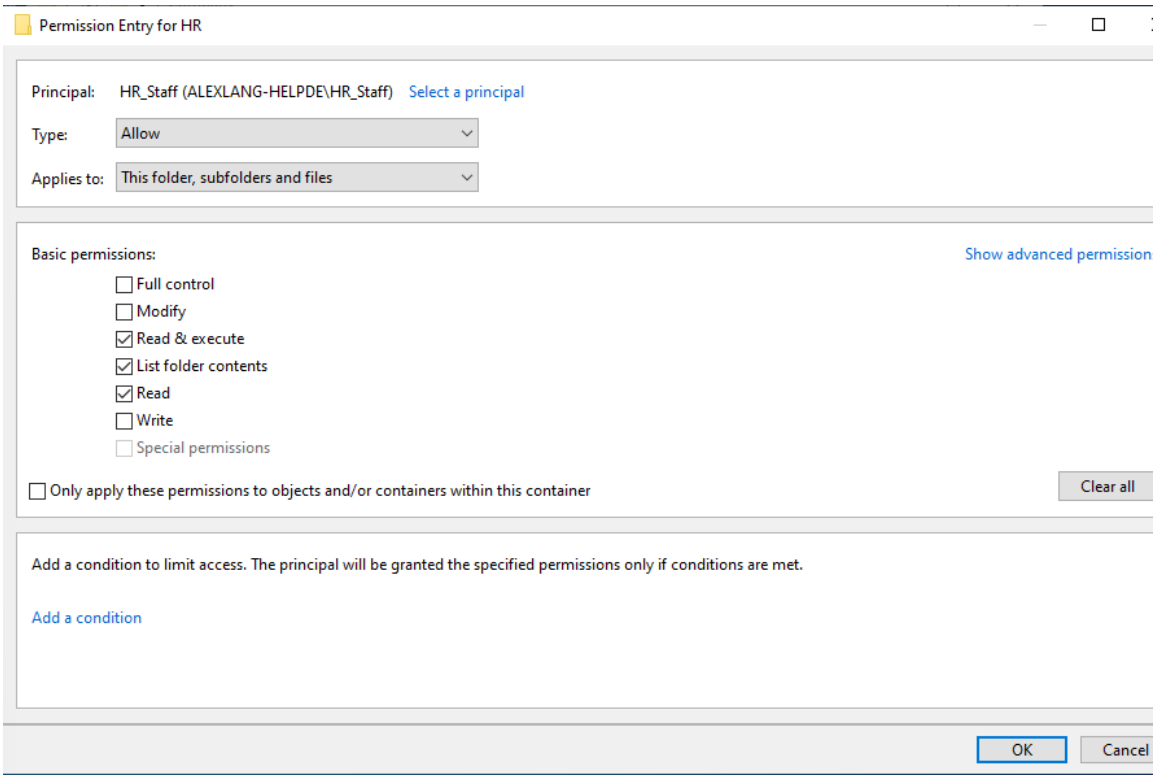
Created a very basic file structure



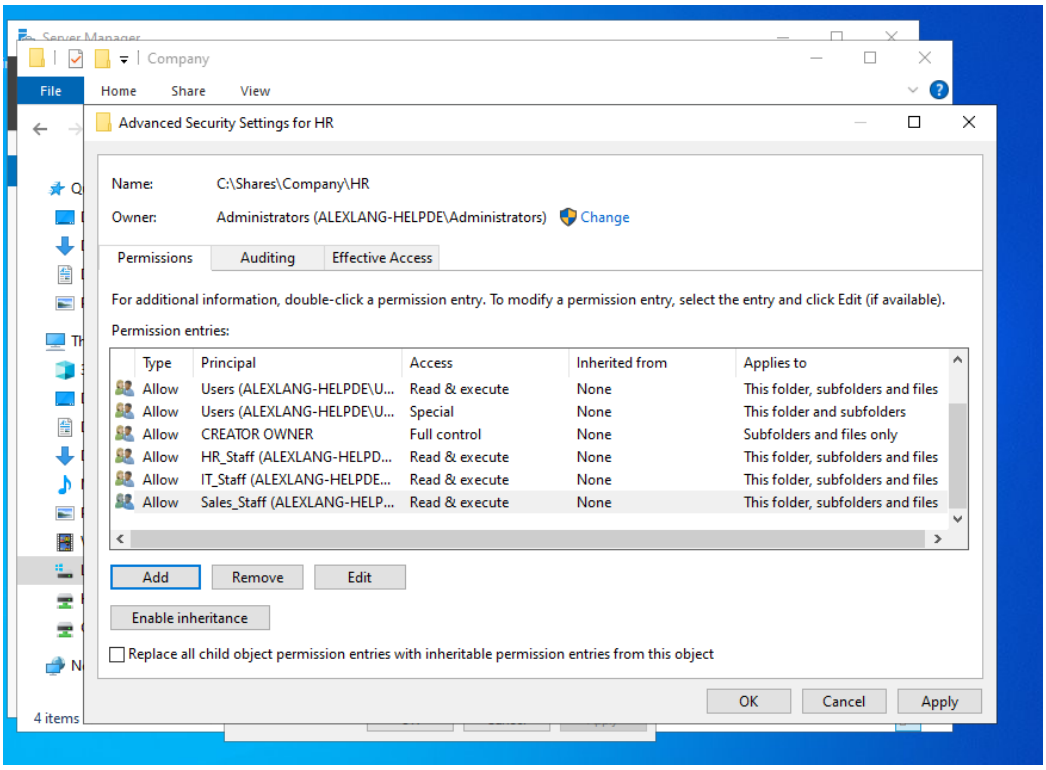
Created 3 new security groups



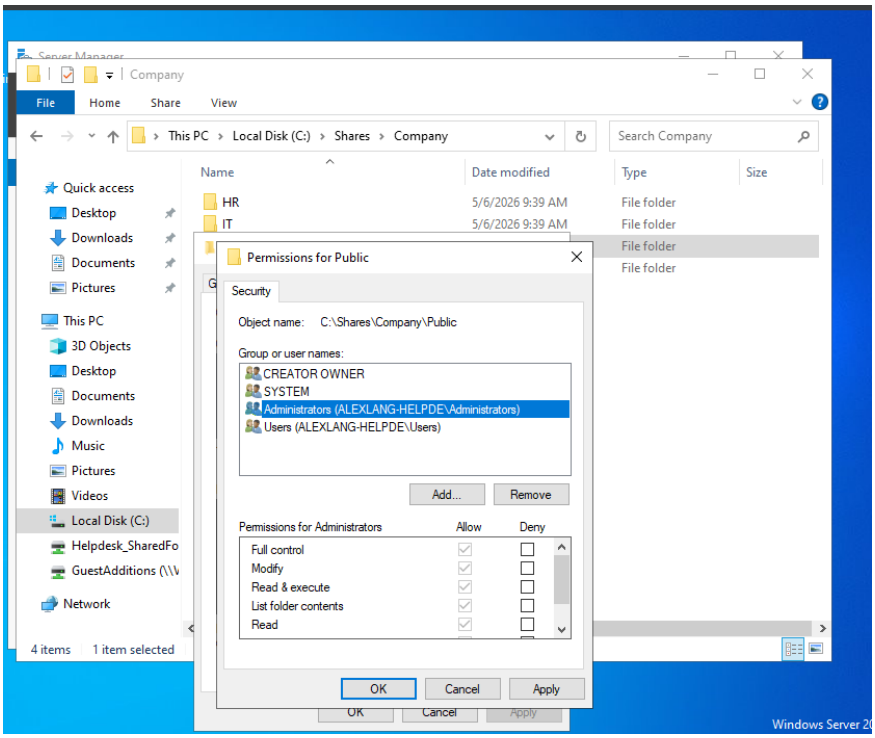
Added approiate users to these groups that have already been created inside of AD.



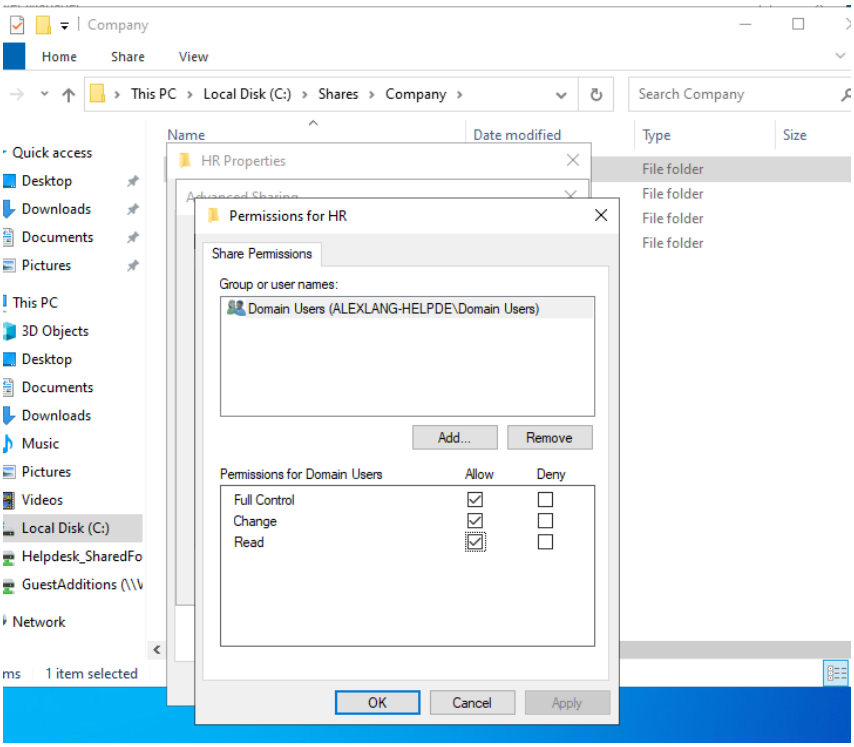
Went into each folder and adjusted inheritance permissions for HR, IT, Sales, and Public



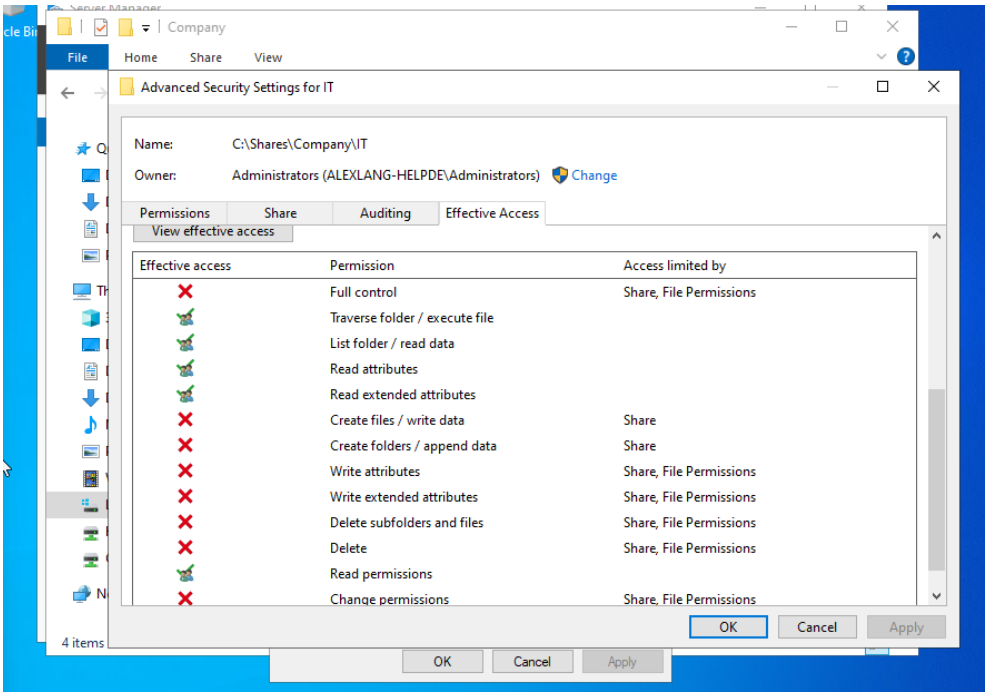
HR, IT, and Sales staff now have read and execute permissions via NFTS permissions



The public folder is supposed to be accessible by everyone, so I made user everyone has read permissions to it

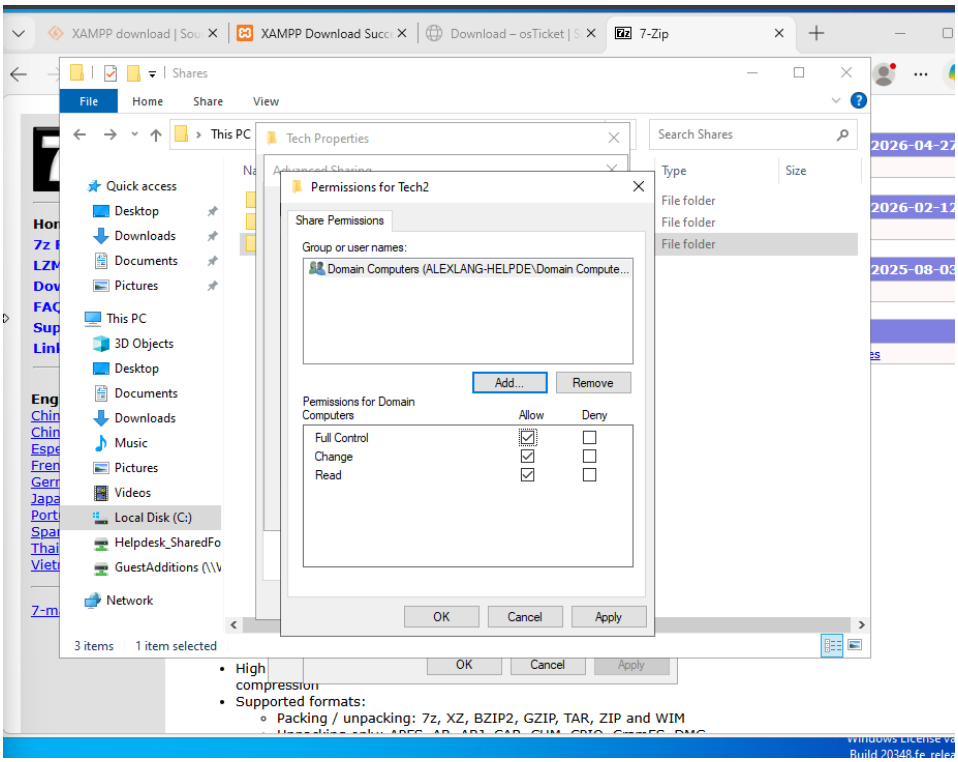


Gave domain users read permissions, and domain admins full control

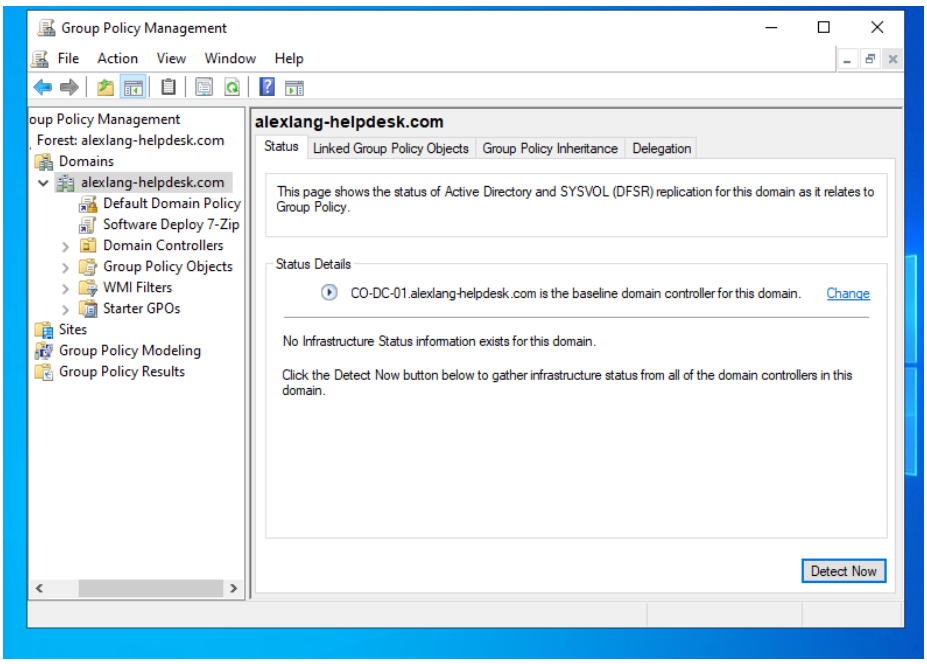


Used effective access control to view all security settings of multiple users to get a full overview of permissions

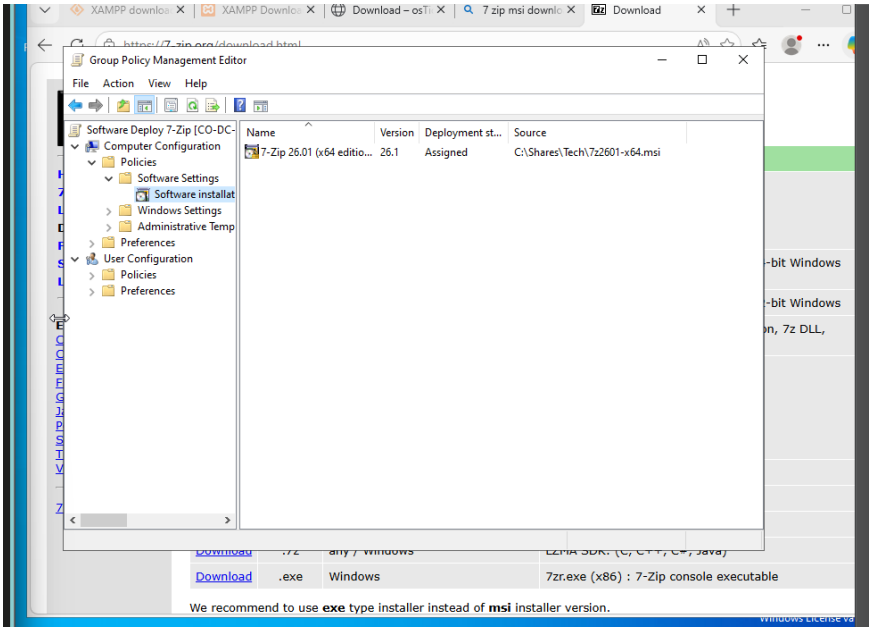
Day 16 Software Deployment through Group Policy



Downloaded 7-Zip on Server controller, and needed to give folder permissions to all domain computers



Created a new group policy object inside of alexlang-helpdesk.com



Package successfully added, ran gpupdate on windows machine, restarted vm, and it auto installed it on that maschine

Observations/Results

- I initially installed the wrong version of 7-ZIP, so I had to go back and install the .msi version instead of the .exe version

Day 17

Observations/Results

Day 18

Observations/Results

Day 19

Observations/Results

Day 20

Observations/Results